

Event & Meeting Room Scheduler

Executive API Audit: Successful Flows, Negative Error Tests, and System Abnormalities

Total Scenarios 30	Successful APIs 26	Validation / Error APIs 4	Abnormalities Identified 5	Overall Test Pass Rate 100%
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1. Codebase Abnormalities, Contract Mismatches & Discrepancies

During deep architectural review against the official assignment specification ('Srilaxmi Assignment: Booking Engine & Conflict Validation Module'), the following discrepancies were identified:

Area	Severity	Description of Abnormality / Discrepancy	Recommended Action
Booking Model Schema	HIGH	Extra Non-Spec Fields: The PostgreSQL schema in the assignment specification only contains: id, room_id, user_id, start_time, end_time, status, created_at. The current model has extra fields: title, description, attendees_count, updated_at.	Align model fields strictly with schema, or make extra fields optional with defaults.
DRF Serializer Payloads	HIGH	Field Name Mismatch (room vs room_id): Serializer expects "room" instead of "room_id". Furthermore, title is marked required in serializer, causing requests following the official spec payload to fail validation.	Accept room_id via PrimaryKeyRelatedField(source='room') and output user_id.
Architecture Pattern	MEDIUM	Missing BookingViewSet: The specification requires 'Implement DRF BookingViewSet endpoints for slot checking and booking creation'. The current implementation uses separate generic views (BookingListCreateView, CheckAvailabilityView, etc.) instead of a unified BookingViewSet(viewsets.ModelViewSet).	Refactor to BookingViewSet with an @action for check-availability, registered on DefaultRouter.
Module Ownership	MEDIUM	Ownership Overlap: /api/my-bookings/ and /api/bookings/{id}/cancel/ belong to Prashanth (User Dashboard & Lifecycle module) per the assignment matrix, but are duplicated in bookings/views.py.	Coordinate clean boundary separation with Prashanth during branch merge.
Database Settings	LOW	SQLite Fallback: settings.py falls back to local SQLite when PostgreSQL credentials are not exported in the shell environment.	Ensure PostgreSQL environment variables are configured in .env.

2. Successful APIs (Positive Flow Verification)

The following 26 endpoints represent all verified normal and authorized flows that responded with successful HTTP status codes (200 OK, 201 Created, 204 No Content):

#	API Name / Purpose	Method	Endpoint Path	Status	Key Validation Verified
1	User 1 Registration	POST	/api/auth/register/	201 Created	Created user Alice with hashed password
2	User 2 Registration	POST	/api/auth/register/	201 Created	Created user Bob in Product dept
4	Admin Authentication	POST	/api/auth/token/	200 OK	Issued admin JWT access/refresh tokens
5	User 1 Authentication	POST	/api/auth/token/	200 OK	Issued user JWT access/refresh tokens
6	Refresh JWT Token	POST	/api/auth/token/refresh/	200 OK	Rotated access token successfully
7	Get Current User Profile	GET	/api/auth/me/	200 OK	Returned authenticated user profile
8	Create Room 1 (Admin)	POST	/api/rooms/	201 Created	Admin created Boardroom Alpha (cap: 20)

#	API Name / Purpose	Method	Endpoint Path	Status	Key Validation Verified
9	Create Room 2 (Admin)	POST	/api/rooms/	201 Created	Admin created Conference Room Beta (cap: 8)
10	List All Rooms	GET	/api/rooms/	200 OK	Returned catalog of active rooms
11	Filter Rooms by Capacity	GET	/api/rooms/?min_capacity=15	200 OK	Filtered rooms having capacity >= 15
12	Filter Rooms by Amenity	GET	/api/rooms/?amenity=Projector	200 OK	Filtered rooms containing Projector
13	Retrieve Room Detail	GET	/api/rooms/{id}/	200 OK	Returned specific room attributes
14	Update Room Detail (Admin)	PATCH	/api/rooms/{id}/	200 OK	Updated location to Executive Wing
15	Check Availability (Free)	GET	/api/bookings/check-availability/	200 OK	Verified is_available = True
16	Create Booking 1	POST	/api/bookings/	201 Created	Confirmed reservation for User 1
18	Check Availability (Busy)	GET	/api/bookings/check-availability/	200 OK	Verified is_available = False with conflicts
19	Create Booking 2 (Room 2)	POST	/api/bookings/	201 Created	Created concurrent booking in Room 2
20	List All Bookings	GET	/api/bookings/	200 OK	Listed active bookings in system
21	Filter Bookings by Room	GET	/api/bookings/?room_id={id}	200 OK	Filtered bookings by room UUID
22	Filter Bookings by Status	GET	/api/bookings/?status=CONFIRMED	200 OK	Filtered only CONFIRMED bookings
23	User Personal Bookings	GET	/api/my-bookings/	200 OK	User dashboard returned Alice's bookings
24	Retrieve Booking Detail	GET	/api/bookings/{id}/	200 OK	Returned single reservation details
25	Update Booking Title	PATCH	/api/bookings/{id}/	200 OK	Updated meeting title successfully
26	Booking Statistics	GET	/api/bookings/stats/	200 OK	Aggregated total/confirmed/cancelled counts
27	Cancel Booking	PATCH	/api/bookings/{id}/cancel/	200 OK	Status transitioned to CANCELLED
29	Check Availability (Freed)	GET	/api/bookings/check-availability/	200 OK	Confirmed slot is freed after cancel
30	Delete Booking	DELETE	/api/bookings/{id}/	204 No Content	Cleaned up record from database

3. Error & Negative Validation APIs (Security & Guardrail Verification)

The following endpoints represent intentional negative validation scenarios testing security, data integrity, and conflict prevention rules:

#	Negative Test Scenario	Method	Endpoint Path	Status Code	Enforced Guardrail & Error Code
3	Duplicate User Registration	POST	/api/auth/register/	400 Bad Request	VALIDATION_ERROR: Rejects duplicate email registration to prevent account takeover.

1 7	Overlapping Room Double-Booking	POST	/api/bookings/	409 Conflict	BOOKING_CONFLICT: Database overlap algorithm detected conflicting slot; returned standard conflict payload.
2 8	Double Cancellation Rejection	PATCH	/api/bookings/{id}/cancel/	400 Bad Request	VALIDATION_ERROR: Rejects repeated cancellation on an already cancelled reservation.
S e c	Unauthenticated Access Guard	GET	/api/rooms/, /api/bookings/	401 Unauthorized	UNAUTHORIZED: Rejects requests missing valid JWT Bearer header.
S e c	Non-Admin Room Creation Guard	POST	/api/rooms/	403 Forbidden	FORBIDDEN: Rejects regular users attempting administrative room modifications.